

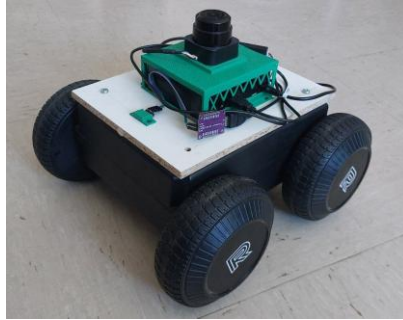
An implementation of Open-RMF in an office environment

Objectives

- At **Elettra Sincrotrone** with the **IT Group**, we wanted to see the Open Robotics Middleware Framework (**Open-RMF**) capacities for use in our warehouse and offices and upgrade our Autonomous Mobile Robots (**AMRs**) to ROS 2 , so we:
 - Build 2 AMRs with ROS 2 and Nav2 creating 2 fleets:
 - One from an UGV **Mini**
 - One porting a **Jobot** AMR from ROS 1 to ROS 2



Mini



Jobot



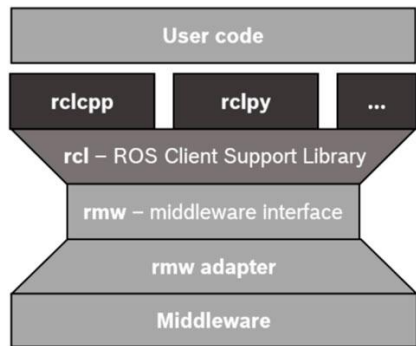
- Port the **DStar** global planner from ROS 1 to ROS 2 to be used in our robots
- And, finally, use **Open-RMF** as middleware for these two fleets



AMR control

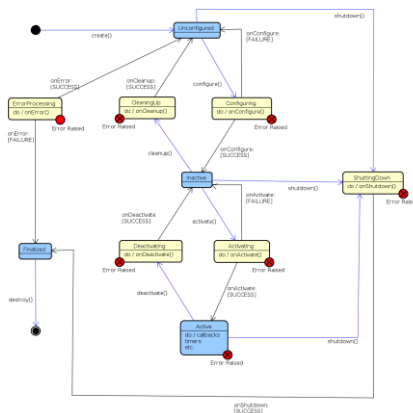
ROS 2 structure and tools

- Communication middleware
- Multiple client libraries : rclcpp, rclpy, rclrs, ...
- **Topics, Service, Action**
- **Lifecycle**
- **Composition**
- **URDF**
- **rosbag2**



Structure

ROS 2

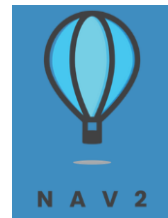
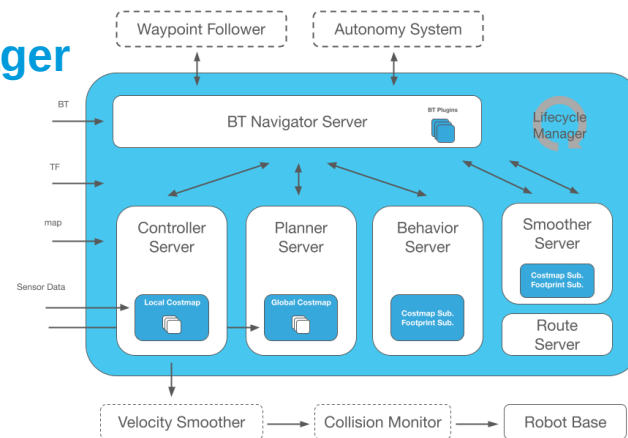


Lifecycle

Nav2 structure

- **Behavior Tree Navigation Server**
- **Planner Server**
- **Smoother Server**
- **Controller Server**
- **Behavior Server**
- **Route Server**
- **Lifecycle manager**

require:
base_link → odom
odom → map



Upgrade from ROS 1 to ROS 2

Upgrade Jobot

- Build system: **catking** → **colcon** (ament_cmake) rclcpp with more C++ 17 features, like smart pointer and new cmake
- **Naming** of variables
- ROS **msgs** update
- **Lifecycle** implemented
 - In **on_configure()** : add creation of secondary nodes (IMU), setup parameters and create callbacks
 - In **on_deactivate()** : cleaning
- Find ros-serial for ROS 2

Start->End	ROS 2 D-start (s)	ROS 2 NavfnPlanner (s)
office -> storage room	1.00	0.020
office -> entrance	5.47	0.150
entrance -> CAD designers	5.36	0.130

Upgrade planner DStar

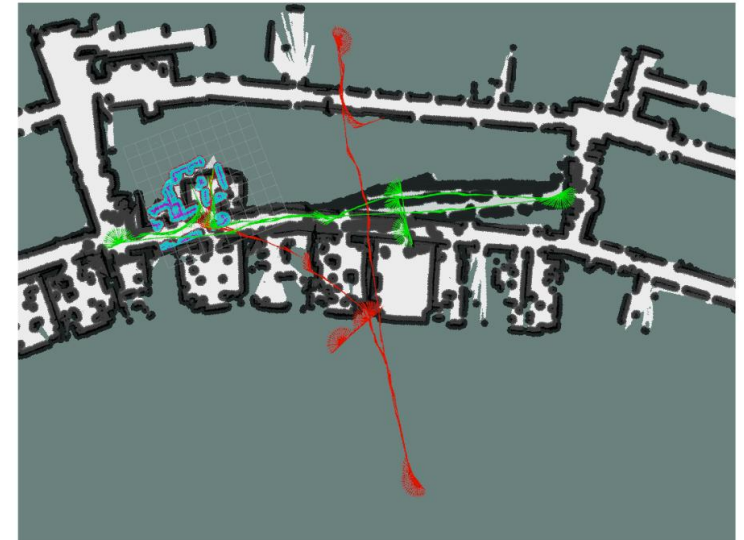
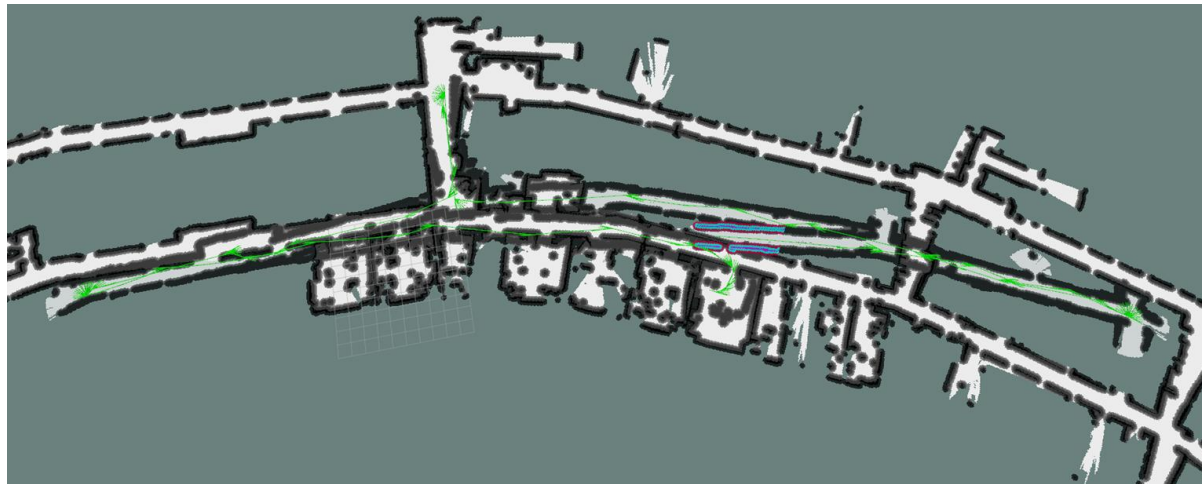
- Inheritance from **nav2_core::GlobalPlanner**
- **Plugin**
- **Lifecycle** structure
 - **configure()** : setup parameters and get pointer
 - **activate()** : fill DStar costmap
 - **cleanup()** : destroy allocate memory
 - **createPlan()** : return a path





robot_localization

- Choosing which **sensor** to fuse
- Calibrating the **process noise covariance matrix Q** of the **EKF**



Global localization

Position respect the map:

SLAM or only **localization**?

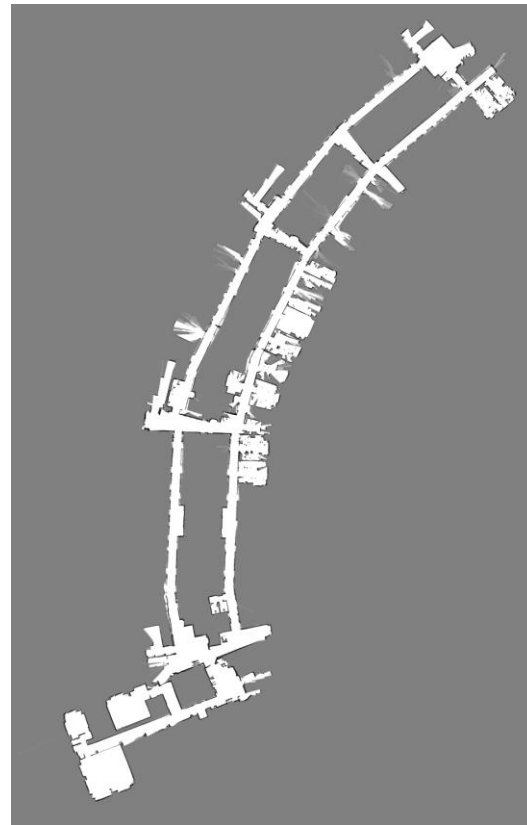
→ Localization:

- **Cartographer**
- **AMCL (Adaptive Monte-Carlo Localizer)**

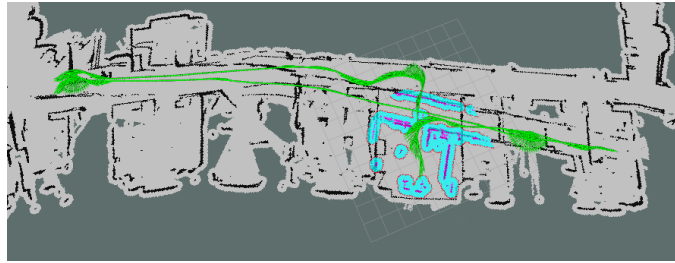
→ Tuning:

→ α parameters:

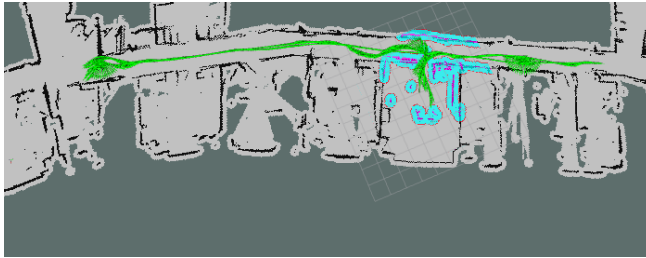
Expected **process noise** in **odometry's**
rotation/translation estimate from
rotation/translation



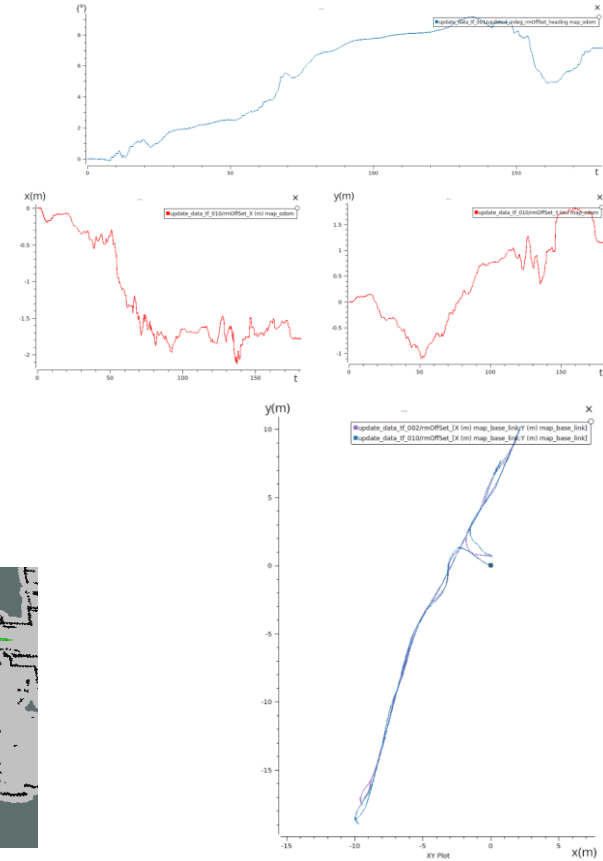
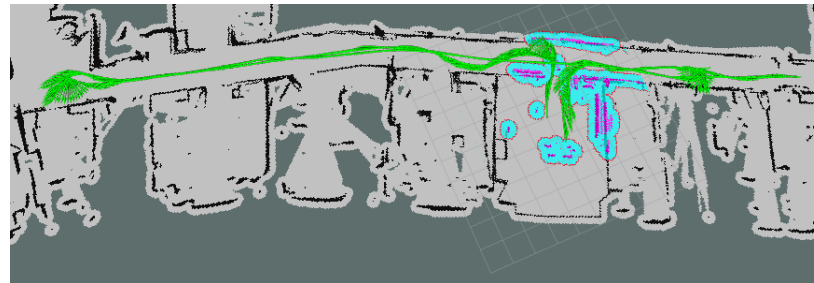
AMCL calibration Mini

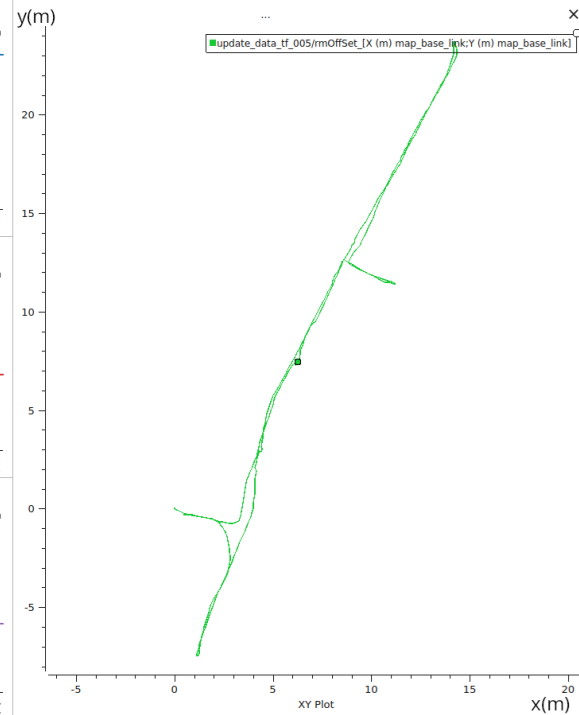
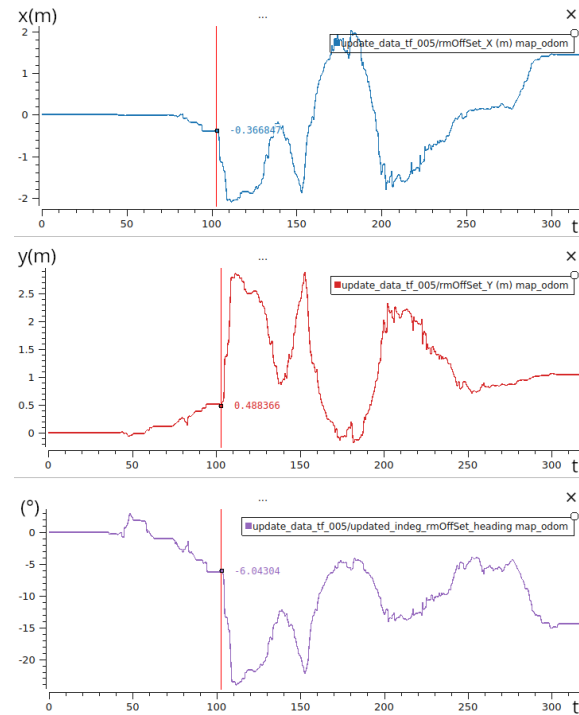
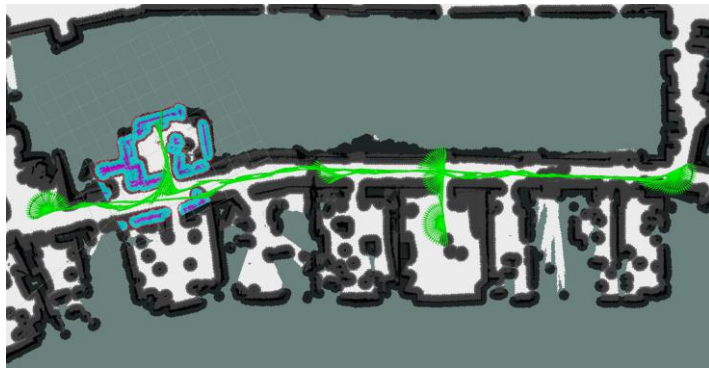
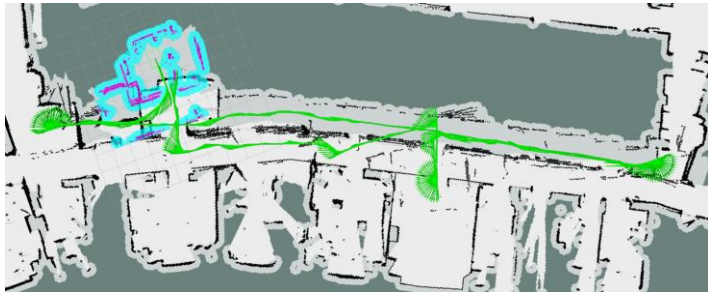


Correcting rotation



Correcting translation

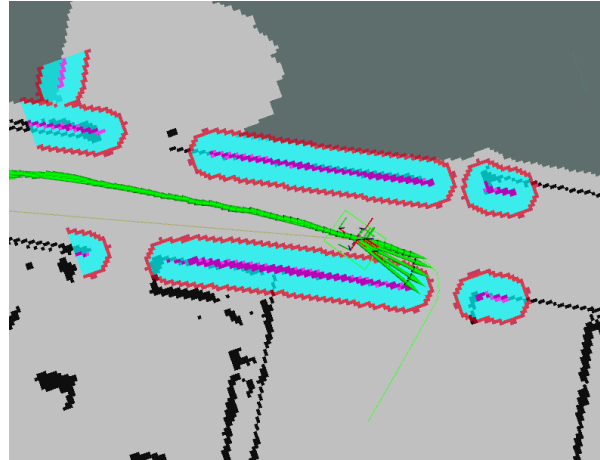




Discovery in simulation



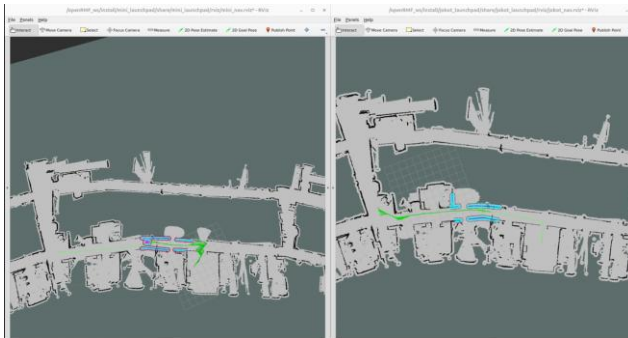
“Error” with the **controller**



DWB
(Dynamic Window Approach)
Controller

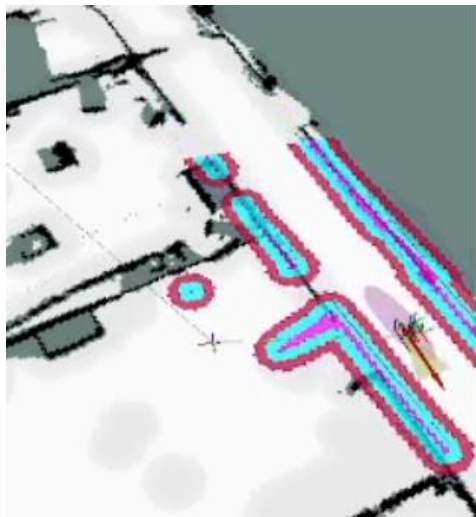


MPPI
(Model Predictive Path Integral)
Controller



Discoveries in reality

- Miss alignment from local costmap and global costmap

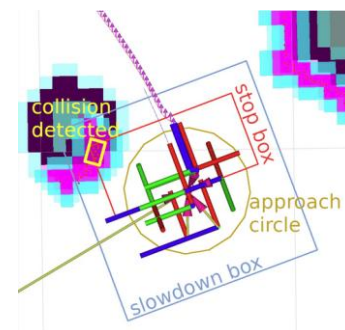
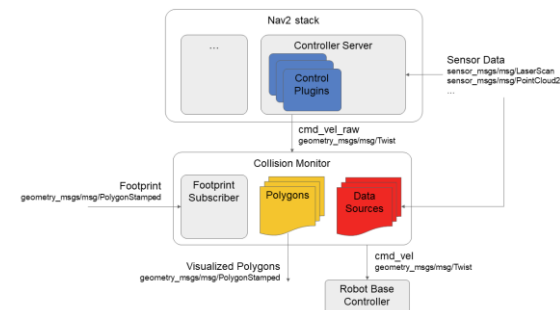


System fail to generate a path to enter the room

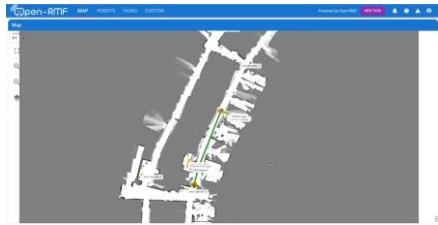
- Obstacles not visible



- Improve safety



Build a system manager for the fleet Mini and for the fleet Jobot

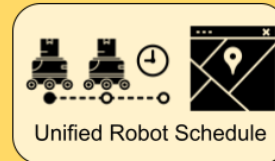


rmf_web

Deployment-specific configuration of
Web UI's and bridges to other IT systems

 Web UI Config 1	WebSocket API
 Web UI Config 2	
⋮	
 Web UI Config N	
 Other System 1	Bridge 1
 Other System 2	Bridge 2
⋮	⋮
 Other System N	Bridge N

**RMF features
common to all
deployments**



Deployment-specific collection of "RMF adapters"
and vendor-provided "managers"

Fleet Adapter 1	Fleet Manager 1	
Fleet Adapter 2	Fleet Manager 2	
⋮	⋮	
Fleet Adapter N	Fleet Manager N or FreeFleet	
Device Adapter 1	Device Manager 1	
Device Adapter 2	Device Manager 2	
⋮	⋮	
Device Adapter N	Device Manager N	

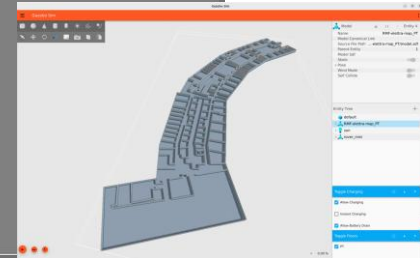
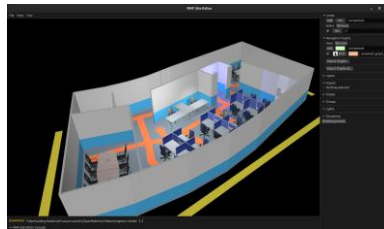
Open-RMF - Navigation graph

rmf_traffic_editor



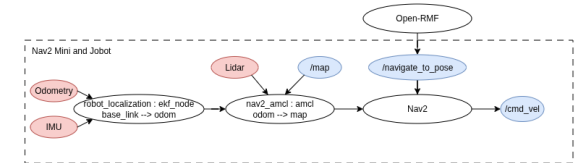
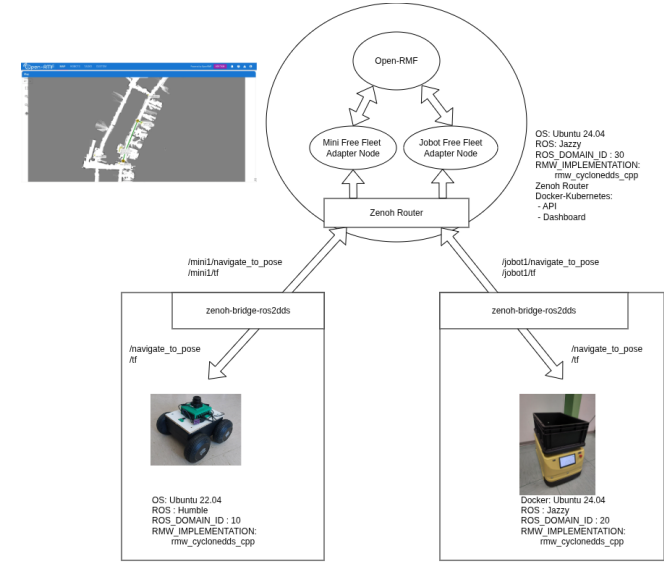
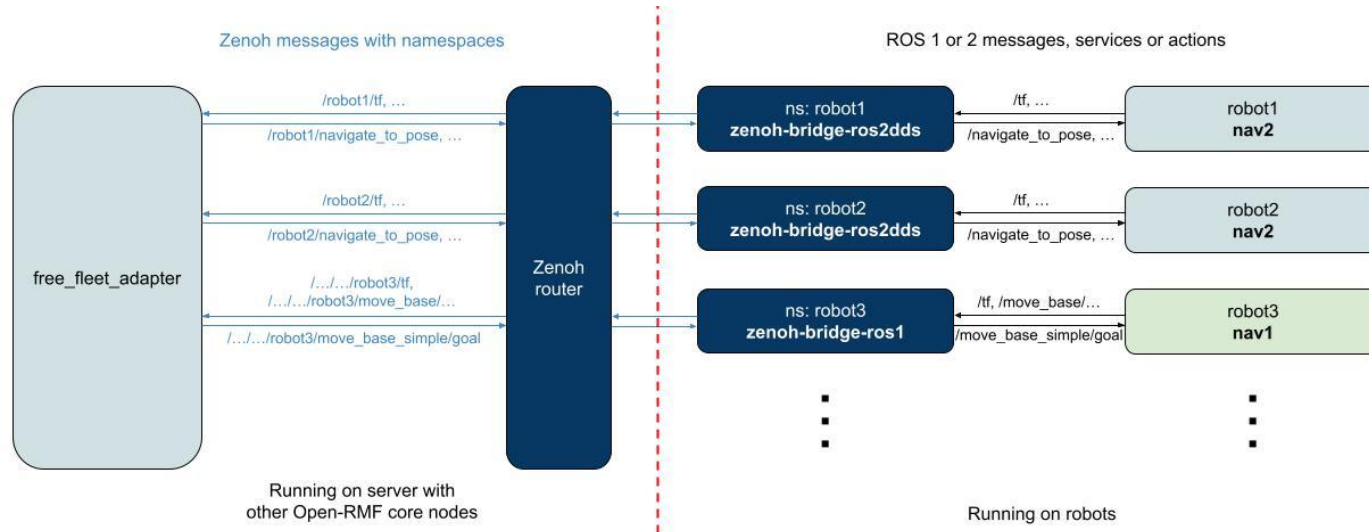
NEW!

rmf_site

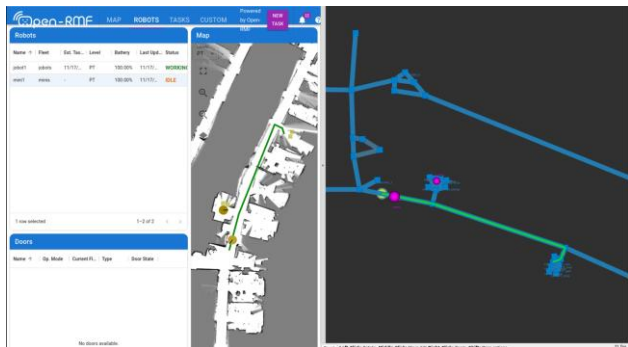


Free Fleet

a ROS 2 dedicated Fleet adapter based on fleet_adapter_template



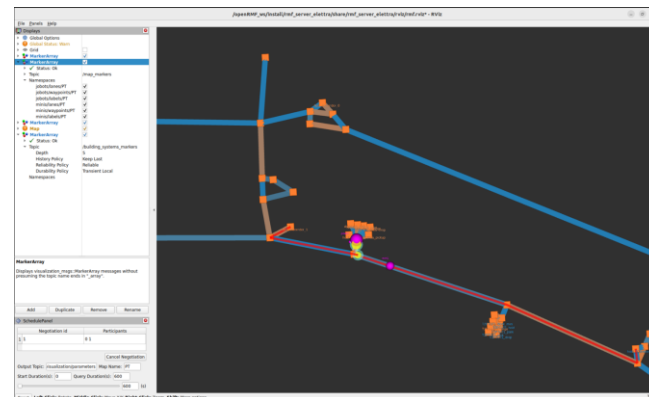
Open-RMF results



Patrol task without negotiation



Mini waiting the Jobot



Patrol task with negotiation

Thanks for the attention!

All the data is available at :

<http://dx.doi.org/10.34965/enodo.i60522>



While the code:

https://gitlab.elettra.eu/open-rmf-tesi/tesi_mag.git

